

Samarth Tiwari

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Education

VIT Bhopal University

B.Tech in Computer Science and Engineering (Cloud Computing and Automation)

Sep 2023 – May 2027

CGPA: 8.45/10

Technical Skills

Languages: Python, Java, JavaScript, SQL

Frontend: HTML5, CSS3

Backend: FastAPI, WebSocket, Flask, REST APIs

Cloud (AWS): Lambda, API Gateway, S3, EC2, CloudWatch, EventBridge, IAM, Bedrock, VPC

Databases: MongoDB, MySQL, DynamoDB

Libraries & Frameworks: NumPy, Pandas, Scikit-learn, TensorFlow/Keras, Matplotlib

Tools & Practices: Docker, Kubernetes, Git, Postman, CI/CD, SDLC

Core CS: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Serverless Computing

Experience

Python Developer – Data Science & AI Intern

May 2025 – July 2025

ExaTip Technologies

Indore, MP

- **Led** end-to-end data pipeline development for **3+ business use cases** — owning preprocessing, EDA, and model delivery — improving dataset quality by **~35%** and cutting model training time via cleaner feature engineering.
- **Built and deployed ML models** (scikit-learn, pandas, NumPy) integrated with REST APIs, achieving **85%+ prediction accuracy** on real-world datasets and enabling live inference for downstream applications.
- **Mentored** a 6-person intern cohort on Git workflows and CI/CD best practices, reducing merge conflicts by **~50%** and improving cross-functional delivery speed across the team.

Projects

Driver Drowsiness Detection

Feb 2026 – Mar 2026

Real-Time Road-Safety AI System

Python, FastAPI, WebSocket, CNN, OpenCV, Docker

- **Built** a real-time CNN-based drowsiness detector (Conv2D 32→32→64 + Softmax) streaming webcam frames at **10 FPS over WebSocket**, achieving **sub-200 ms** inference latency — cutting false-negative safety alerts by **~30%** via confidence-threshold fallback logic.
- **Containerized** the full-stack app (FastAPI backend + Vercel frontend) with Docker and published to Docker Hub; reduced deployment setup from **hours to under 2 minutes** across any cloud or local environment.
- **Boosted low-light detection accuracy by ~25%** using OpenCV Haar Cascade ROI extraction with Histogram Equalization, ensuring reliable performance across real-world lighting conditions.

Packet Analyser

Jan 2026 – Mar 2026

High-Performance Deep Packet Inspection (DPI) Engine

Python, Computer Networks

- **Engineered** a **multi-threaded DPI pipeline** with consistent hashing and lock-safe queues, enabling **zero-race concurrent** packet processing with accurate stateful flow tracking across high-volume traffic.
- **Reduced** traffic classification latency by **~40%** via TLS SNI extraction and five-tuple flow modeling, enabling real-time application identification and automated policy enforcement at scale.

Sahayak AI (Live)

Sept 2025 – Jan 2026

An AI-Powered Classroom Assistant for Indian Middle Schools

AWS Lambda, Bedrock, DynamoDB, S3, Python

- **Automated** end-to-end assignment evaluation pipeline, reducing manual grading effort by **~80%** by processing Google Forms/Doc submissions via **Lambda** and **Bedrock**.
- Scaled serverless architecture handling structured/unstructured responses, enabling real-time grading and feedback delivery with **DynamoDB** storage and email notifications.
- Enabled data-driven decision making by **aggregating student performance data** and powering dashboard insights QuickSight for class-level and individual tracking.

Achievements & Certifications

- **IBM** Professional Data Scientist
- **Oracle** Professional Data Scientist and OCI Foundation Associate
- **Winner** – Nebula Nexus Hackathon 2025, MIT Jaipur